



Effect of supplementing live *Saccharomyces cerevisiae* yeast on performance, rumen function, and metabolism during the transition period in Holstein dairy cows

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ABSTRACT

Dairy cows have to face several nutritional challenges during the transition period, and live yeast supplementation appears to be beneficial in modulating rumen activity. In this study, we evaluated the effects of live yeast supplementation on rumen function, milk production, and metabolic and inflammatory conditions. Ten Holstein multiparous cows received either live *Saccharomyces cerevisiae* (strain Sc47; SCY) supplementation from –21 to 21 d from calving (DFC) or a control diet without yeast supplementation. Feed intake, milk yield, and rumination time were monitored until 35 DFC, and rumen fluid, feces, milk, and blood samples were collected at different time points. Compared with the control diet, SCY had increased dry matter intake (16.7 vs. 19.1 ± 0.8 kg/d in wk 2 and 3) and rumination time postpartum (449 vs. 504 ± 19.9 min/d in wk 5). Milk yield tended to be greater in SCY (40.1 vs. 45.2 ± 1.7 kg/d in wk 5), protein content tended to be higher, and somatic cell count was lower. In rumen fluid, acetate molar proportion was higher and that of propionate lower at 21 DFC, resulting in increased acetate:propionate and (acetate + butyrate):propionate ratios. Cows in the SCY group had lower fecal dry matter but higher acetate and lower propionate proportions on total volatile fatty acids at 3 DFC. Plasma analysis revealed a lower degree of inflammation after calving in SCY (i.e., lower haptoglobin concentration at 1 and 3 DFC) and a likely better liver function, as suggested by the lower γ -glutamyl transferase, even though paraoxonase was lower at 28 DFC. Plasma IL-1 β concentration tended to be higher in SCY, as well as Mg and P. Overall, SCY supplementation improved rumen and hindgut fermentation profiles, also resulting in higher dry mat-

ter intake and rumination time postpartum. Moreover, the postcalving inflammatory response was milder and liver function appeared to be better. Altogether, these effects also led to greater milk yield and reduced the risk of metabolic diseases.

Key words: nutraceutical, rumen function, probiotic, health

INTRODUCTION

The transition period is recognized as the most challenging phase of the life of a dairy cow. At the turn of the calving event, many metabolic, immune, and endocrine changes happen, and physiological imbalances take place (Drackley, 1999; Ingvarsen and Moyes, 2013; Trevisi and Minuti, 2018). Energy and nutrient requirements dramatically increase, whereas potential feed intake is limited, resulting in a negative energy balance (Grummer et al., 2004). The inflammatory condition typical of the transition period contributes to the reduction of feed intake and daily rumination time (Calamari et al., 2014). Moreover, the abrupt change in the energy content of the diet requires the rumen to adapt, with possible detrimental effects on cow performance when adaptation fails (Jouany, 2006; Bannink et al., 2012; Steele et al., 2015). Rumen is involved in the inflammatory and oxidative stress responses around calving, which are partially consequences of the modifications in rumen microbiota and fermentation patterns (Minuti et al., 2015; Trevisi et al., 2018).

Among the several nutritional strategies proposed to cope with transition period challenges (Lopreiato et al., 2020), supplementation with yeast products became widespread, despite variable results, depending on dose, strain, and type of yeast product used, and on the diet and physiological condition of receiving animals (Chaucheyras-Durand et al., 2008; Robinson and Erasmus, 2009; Allen and Ying, 2012). Although the mode of action of these products has yet to be

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fully elucidated, positive effects observed in mature ruminants on health, efficiency, and performance can be mainly related to the stimulus to rumen microbiome, the stabilization of ruminal pH, the improved fiber degradation in the rumen (Chaucheyras-Durand et al., 2008; Burdick Sanchez et al., 2021), and the interactions with gastrointestinal epithelium (Filipe et al., 2019). Different types of yeast products have been tested in ruminants over the years, both as live and dead forms (e.g., yeast cell walls or fermentation products). *Saccharomyces cerevisiae* represents one of the most used strains in ruminant nutrition, with benefits on intake, rumen fermentation, and milk production (Desnoyers et al., 2009), even though DMI was not always increased (Bach et al., 2007; DeVries and Chevau, 2014). Supplementation with live *S. cerevisiae* mitigated the effects of the negative energy balance, with a decreased lactate accumulation in the rumen fluid, paired with increased rumen pH and reduced blood nonesterified fatty acids and BHB (Kumprechtová et al., 2019). Moreover, supplementation of live *S. cerevisiae* increased the expression of genes regulating inflammation and epithelial barrier integrity in the rumen, both before and after calving (Bach et al., 2018), and altered rumen microbiota before calving (Bach et al., 2019).

We hypothesized that supplementing live yeast during the transition period could ameliorate inflammatory condition of periparturient cows and improve rumen function at the onset of lactation. Thus, this study aimed to evaluate the effects of *S. cerevisiae* supplemented in viable form during transition period on rumen function, milk production, and metabolic and inflammatory conditions.

MATERIALS AND METHODS

Animal Management and Experimental Design

The study was performed between September 2017 and May 2018 at the Università Cattolica del Sacro Cuore research dairy barn (Experiment Station, San Bonico, Piacenza, Italy) in accordance with Italian laws

on animal experimentation (DL n. 26, 04/03/2014) and ethics (Authorization of Italian Health Ministry N 1047/2015-PR). A group of 10 multiparous Holsteins dairy cows [parity: 2.7 ± 0.7 ; BW: 662 ± 61 kg; milk yield in the last lactation: $10,526 \pm 879$ kg (mean $\pm$ SD)] was raised in individual tiestalls with controlled environmental conditions (room temperature of 20°C, relative humidity of 65%, 14 h of light) from -35 to 35 d from calving (DFC). Cows were transferred to the barn before the dry-off and were individually fed a component diet. They were fed with 2 meals of forages (grass hay and corn silage during dry period; grass hay, alfalfa hay, and corn silage during lactation) offered at intervals of 12 h, and 2 to 8 meals of concentrates at equal intervals (2 in dry period and 4 to 8 in lactation according to the DIM and milk production) distributed by an automatic feeder. The amount of concentrate was gradually increased after calving by 0.5 kg/d until energy and protein requirements were covered (1 kg of concentrate every 3 kg of milk produced). Vitamin and mineral supplements were added to corn silage twice a day. Feeds offered and refused by cows were weighed daily.

Animals were blocked by parity and randomly assigned to the following 2 groups ($n = 5/\text{group}$): control group (CTR) and yeast-supplemented group (SCY). Cows in SCY group were supplemented with 10 g/d of live yeast (*S. cerevisiae*, ActiSaf Sc47, 1×10^{10} cfu/g, Phileo by Lesaffre, Marcq-en-Baroeul, France) once a day from -21 d before the expected calving date until 21 DFC. Study personnel was not blind to treatment groups. Supplement was stored in bags in the dark and at room temperature. Before administration, it was mixed with 100 g of corn silage and added to the feed bin every morning, on top of forage meal. Full intake of the supplement was checked every day. Likewise, CTR cows received 100 g of corn silage every day without any supplementation. Refusals were removed and weighed every day, immediately before the addition of new meal. Representative samples of feeds were collected during the trial for the determination of DM and chemical composition (AOAC International, 2012), shown in Table 1.

Table 1. Chemical composition of feeds offered to the cows during the trial (expressed as mean $\pm$ SD)

Item	Corn silage	Dehydrated alfalfa	Grass hay	Concentrate
DM, %	36.2 ± 1.38	94.2 ± 0.42	90.8 ± 0.36	89.1 ± 1.65
NDF, % of DM	38.0 ± 1.48	40.6 ± 1.51	58.0 ± 1.16	17.5 ± 2.48
ADF, % of DM	23.1 ± 0.76	29.8 ± 3.00	37.4 ± 0.46	9.46 ± 0.62
ADL, % of DM	2.45 ± 0.19	6.59 ± 0.48	5.23 ± 0.63	1.47 ± 0.32
Starch, % of DM	32.1 ± 0.91	—	—	28.3 ± 3.19
CP, % of DM	8.86 ± 0.71	19.9 ± 1.65	9.64 ± 0.65	20.1 ± 0.94
Ash, % of DM	4.14 ± 0.23	9.51 ± 0.69	8.12 ± 1.11	7.39 ± 0.42

DMI, BCS, Rectal Temperature, and Health Status

Dry matter intake was calculated daily by weighing the amount of feed offered and refused by each cow and converted into a DM basis according to the DM of the individual feeds. Body weight was measured at −28, −14, 1, 14, and 28 DFC, and BCS was assessed at the same time points as BW with a 4-point scale (Agricultural Development and Advisory Service, 1986) by the same operator. Rectal temperature was measured twice a week throughout the trial, and it was expressed as weekly average. Health status was checked daily by farm veterinarians and study personnel. Diseases and their corresponding detection dates were registered.

Blood Collection and Analysis

At −21, −7, 1, 3, 7, 21, and 28 DFC, blood samples were collected from jugular vein into evacuated heparinized tubes (BD Vacutainer; BD and Co.) before the morning feeding, and processed as described by Calamari et al. (2016). After collection, samples were immediately placed in an ice-water bath. Then, they were centrifuged at $3,500 \times g$ for 16 min at 4°C, and packed cell volume was determined. A clinical auto-analyzer (ILAB-650, Instrumentation Laboratory Werfen) was used to determine concentrations of glucose, BHB, non-esterified fatty acid, urea, creatinine, cholesterol, total protein, albumin, globulin, haptoglobin (Skinner et al., 1991), ceruloplasmin (Sunderman and Nomoto, 1970), paraoxonase (Ferré et al., 2002), γ -glutamyl transferase (GGT), aspartate aminotransferase, bilirubin, Ca, P, Mg, Zn, reactive oxygen metabolites, myeloperoxidase (Bradley et al., 1982), advanced oxidation protein products (Hanasand et al., 2012), ferric reducing antioxidant power (Benzie and Strain, 1999), and thiol groups (Supplemental Table S1; <https://doi.org/10.6084/m9.figshare.21756044>; Cattaneo et al., 2023). From samples collected at −21, −7, 3, and 21 DFC, concentrations of IL-1 β and IL-6 were analyzed by commercial bovine ELISA kits (Thermo Fisher Scientific), as described by Mezzetti et al. (2019).

Rumination Time, Rumen Fluid, and Fecal Sampling and Analysis

Rumination time was recorded daily with Hr-LD tags (SCR by Allflex) from −35 to 35 DFC. Data were expressed as a weekly average. At 21 DFC, rumen fluid was collected 6 h after the morning forage feeding using a ruminal probe specially designed for cattle (Ruminator; profs-products.com), as described by Wallace et al. (2019). To avoid salivary contamination, the first liter of collected fluid was discarded, and the next 0.5

L was retained. For all samples, pH was recorded immediately with a pH meter (GLP 21; Crison Instruments SA). Afterward, samples were gently mixed, and 5 aliquots of 1 mL each were pipetted into 2-mL tubes and immediately stored at −20°C for subsequent VFA measurement.

At −21, −7, 3, and 21 DFC, fecal samples were collected manually in plastic bags from rectal ampulla immediately after rumen sampling. For all samples, pH was determined on fresh material after mixing, as the mean of 6 readings at different points of the sample. For VFA analysis, 10 g of fecal sample was added to 90 mL of distilled water and homogenized for 2 min at maximum speed (260 rpm) in a Stomacher (Seward Stomacher 400 Circulator, International PBI). The homogenate aliquots were stored at −20°C until analysis. Fecal DM was determined from 200 g of fresh material after 96 h in a ventilated oven at 65°C.

Concentrations VFA in rumen fluid (−21 DFC) and fecal homogenate (−21, −7, 3, and 21 DFC) were analyzed by a gas chromatograph (model 7820A; Agilent Technologies), and urea, D- and L-lactic acid in rumen fluid as described by Minuti et al. (2014). Data about individual VFA were reported as molar proportions of the total VFA concentration. Urea was measured using a urea nitrogen kit (Instrumentation Laboratory Werfen) with a clinical auto-analyzer (ILAB-650), and D- and L-lactic acid were determined with a D-/L-Lactate assay kit (Megazyme International Ltd.), following manufacturer's instructions.

Milk Yield, Sampling, and Composition Assessment

Up to 35 DFC, milk yield was weighed at each milking session in the parlor and expressed as a weekly average. At 7, 14, 21, and 28 DFC, approximately 100 mL of milk was collected for the determination of milk composition and SCC. Milk composition (fat, protein, lactose, casein, urea) was determined using a Milkoscan FT120 analyzer (Foss Analytics), and SCC by Fossomatic 180 (Foss Analytics).

Statistical Analysis

Sample size was determined according to logistical and financial constraints due to the intense sampling and analytical protocol, also considering that cows were kept in a controlled environment (constant ambient temperature, relative humidity, and photoperiod). Data were analyzed with SAS software (release 9.3, SAS Institute Inc.). Normal distribution of data was checked by using UNIVARIATE procedure in SAS. Not normally distributed data were log-transformed and presented as back-transformed data. Data about

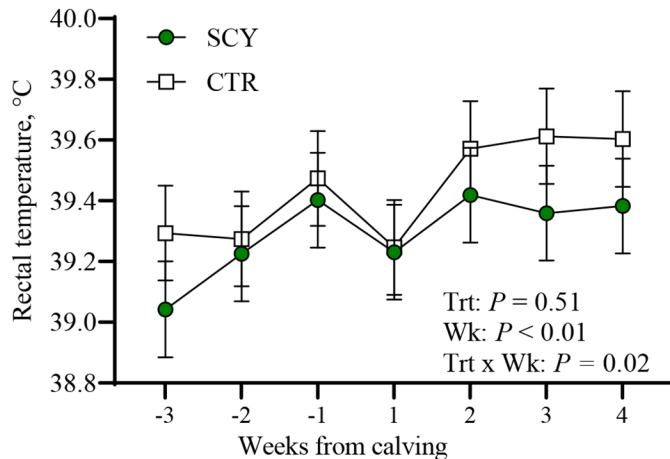


Figure 1. Rectal temperature from -3 to 3 wk from calving in Holstein dairy cows fed a diet not supplemented (CTR) or supplemented with live yeast *Saccharomyces cerevisiae* Sc47 (10 g/d; SCY) from -21 to 21 d relative to calving. Data are presented as LSM $\pm$ SEM. *P*-values of the main effect are reported in the figure. Trt = treatment; Wk = week.

rumen and fecal samples, milk, feeding, and rumination behavior were subjected to ANOVA with repeated measures using the MIXED procedure of SAS. Data about plasma biomarkers were analyzed with repeated measures mixed models using the GLIMMIX procedure. The statistical models included the fixed effects of treatment, time (days from calving), and their interaction, and the random effect of individual cow. Four structures of covariance (compound symmetry, autoregressive order, Toeplitz, or spatial power) were tested and the one with the lowest Akaike information criterion was retained in the model. Pre-experimental checks were included as a covariate when appropriate. Pair-wise comparisons were performed using the least significant difference test. Statistical significance was declared at $P \leq 0.05$ and tendencies at $P \leq 0.10$.

RESULTS

Health Status, BW, and BCS

During the dry period, no clinical disease was observed. After calving, there was a case of retained placenta in CTR. In CTR, we also noted 1 case of retarded uterine involution (at 11 DFC) and 1 subclinical ketosis (at 10 DFC). The SCY group only showed a case of subclinical ketosis (at 11 DFC). None of the problems were severe. Despite overall similar values (39.3 vs. $39.4 \pm 0.15^\circ\text{C}$ in CTR and SCY, respectively; $P = 0.52$), rectal temperature had numerically lower values after calving in SCY compared with CTR, resulting in a treatment (Trt) $\times$ time interaction effect ($P = 0.03$;

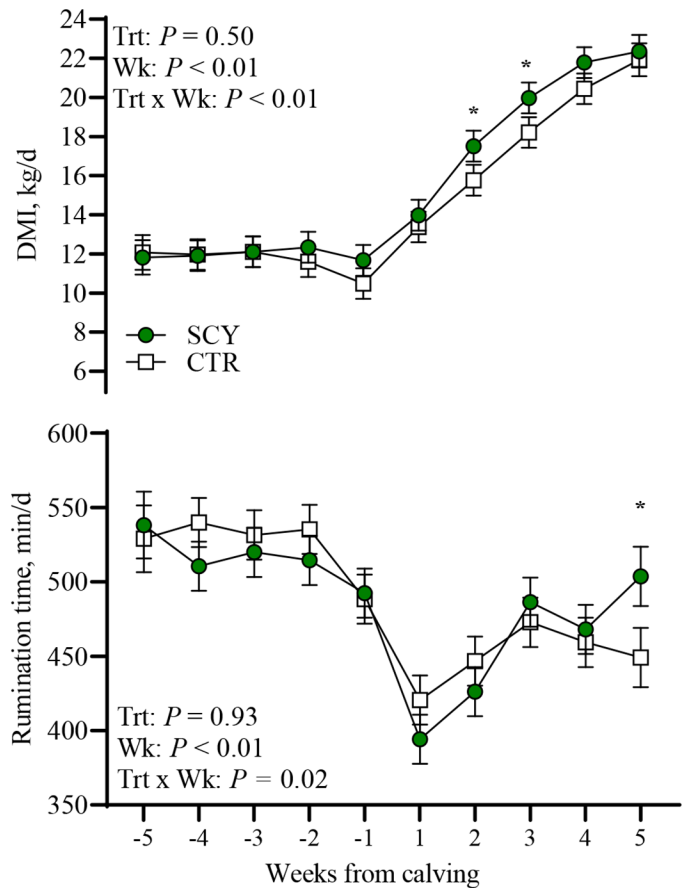


Figure 2. Dry matter intake and daily rumination time from -5 to 5 wk from calving in Holstein dairy cows fed a diet not supplemented (CTR) or supplemented with live yeast *Saccharomyces cerevisiae* Sc47 (10 g/d; SCY) from -21 to 21 d relative to calving. Data are presented as LSM $\pm$ SEM. *P*-values of the main effect are reported in the figure, and differences between groups at each time point are denoted with *. Trt = treatment; Wk = week.

Figure 1). No differences were observed in BW (613 vs. 640 ± 27.1 kg; Trt, $P = 0.50$) and BCS (2.36 vs. 2.31 ± 0.08 ; Trt, $P = 0.63$).

Feeding and Rumination Behavior

Daily DMI and rumination time are shown in Figure 2. No differences were observed between groups before calving, whereas SCY tended to have a greater DMI in wk 2 and 3 after calving (15.8 and 18.2 ± 0.78 kg/d in CTR vs. 17.5 and 20.0 ± 0.78 kg/d in SCY, respectively; $P = 0.10$), resulting in a significant interaction effect of Trt $\times$ time ($P < 0.01$). Average rumination time did not differ between groups ($P = 0.93$), but a greater increase in the SCY group compared with CTR at wk 5 (449 vs. 504 ± 19.9 min/d in CTR and SCY, respectively) resulted in a significant interaction effect of Trt $\times$ time ($P = 0.02$).

Table 2. Rumen fluid pH, lactate, urea, and VFA at 21 d after calving of Holstein dairy cows fed a diet not supplemented (CTR) or supplemented with live yeast *Saccharomyces cerevisiae* Sc47 (10 g/d; SCY) from –21 to 21 d relative to calving

Item	Treatment		SEM ¹	P-value
	CTR	SCY		
pH	6.57	6.77	0.14	0.35
D-Lactate, mmol/L	41.8	58.6	13.7	0.41
L-Lactate, mmol/L	43.0	58.7	13.6	0.44
Urea, mg/L	1.92	2.70	0.57	0.36
Total VFA, mmol/L	91.3	81.3	8.86	0.45
Individual VFA, mol/100 mol				
Acetate	57.6	63.0	1.68	0.05
Propionate	26.8	20.6	2.20	0.08
Butyrate	12.0	12.6	1.04	0.70
Isobutyrate	0.67	0.93	0.15	0.26
Valerate	1.47	1.21	0.14	0.25
Isovalerate	1.02	1.31	0.20	0.33
Hexanoate	0.42	0.35	0.05	0.41
Heptanoate	0.04	0.02	0.01	0.25
Acetate:propionate ratio	2.27	3.16	0.31	0.08
(Acetate + butyrate):propionate ratio	2.75	3.80	0.39	0.09

¹Greatest SEM.

Rumen pH, D-lactate, L-lactate, urea, and total VFA production did not differ between groups (Table 2). However, SCY cows tended to have increased acetate:propionate and (acetate + butyrate):propionate ratios ($P = 0.08$ and $P = 0.09$, respectively). The latter was the result of the increased acetate and decreased propionate proportions ($P = 0.05$ and $P = 0.08$, respectively). Other VFA proportions did not differ between groups.

In feces (Table 3), DM tended to be lower in SCY compared with CTR at 3 DFC (Trt $\times$ time; $P = 0.07$; Figure 3), whereas pH and total VFA content were not different. At 3 DFC, acetate tended to be higher (72.7 vs. 74.5 ± 0.6 mol/100 mol in CTR and SCY, respectively; $P = 0.07$) and propionate was lower (17.9 vs. 15.8 ± 0.6 mol/100 mol; $P = 0.03$) in SCY versus CTR,

whereas at 21 DFC acetate was lower (75.5 vs. 73.7 ± 0.6 mol/100 mol; $P = 0.05$). Similarly, the acetate:propionate ratio was higher at 3 DFC and lower at 21 DFC in the SCY group compared with CTR (Trt $\times$ time; $P = 0.05$; Figure 3).

Milk Production

Average milk yield in the first 5 wk of lactation is shown in Figure 4. Compared with CTR, SCY group had higher production in wk 5, leading to a tendency toward an interaction of Trt $\times$ time effect (40.1 vs. 45.2 ± 1.7 kg/d in CTR and SCY, respectively; $P = 0.08$). Milk composition is reported in Table 4. Milk protein tended to be higher in SCY group (Trt $\times$ time; $P =$

Table 3. Fecal DM, pH, and VFA of Holstein dairy cows fed a diet not supplemented (CTR) or supplemented with live yeast *Saccharomyces cerevisiae* Sc47 (10 g/d; SCY) from –21 to 21 d relative to calving

Item	Treatment			P-value		
	CTR	SCY	SEM ¹	Trt ²	Day	Trt $\times$ day
DM, %	13.6	13.0	0.97	0.44	0.46	0.07
pH	6.62	6.65	0.12	0.73	<0.01	0.88
Total VFA, mmol/L	4.15	3.86	0.47	0.31	<0.01	0.27
Individual VFA, mol/100 mol						
Acetate	75.0	74.7	0.91	0.56	<0.05	0.07
Propionate	16.0	15.6	0.89	0.58	<0.05	0.05
Butyrate	5.87	6.21	0.62	0.43	<0.01	0.99
Isobutyrate	1.13	1.27	0.15	0.24	<0.01	0.18
Valerate	1.21	1.24	0.08	0.51	<0.01	0.80
Isovalerate	0.81	0.93	0.18	0.36	<0.01	0.59
Acetate:propionate ratio	4.78	4.81	0.29	0.88	<0.05	0.05
(Acetate + butyrate):propionate ratio	16.4	15.6	1.79	0.56	<0.01	0.94

¹Greatest SEM.²Trt = treatment.

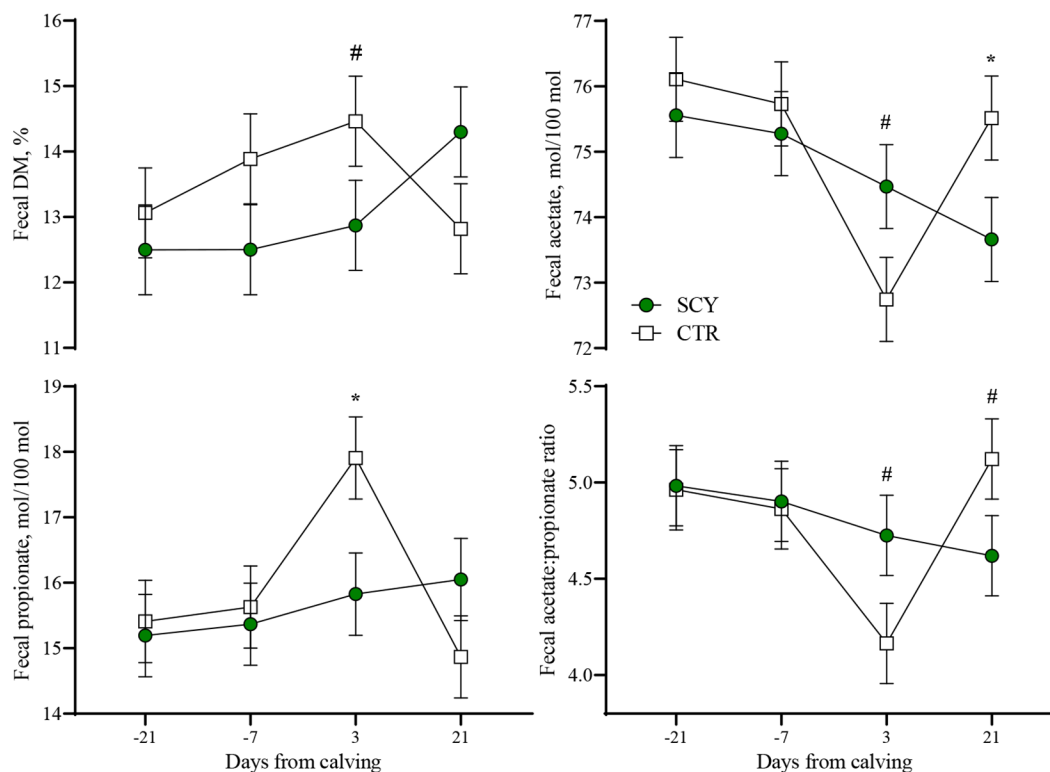


Figure 3. Fecal DM, acetate, propionate, and acetate:propionate ratio in Holstein dairy cows fed a diet not supplemented (CTR) or supplemented with live yeast *Saccharomyces cerevisiae* Sc47 (10 g/d; SCY) from -21 to 21 d relative to calving. Data are presented as LSM $\pm$ SEM. Differences between groups at each time point are denoted with * ($P < 0.05$) or # ($P < 0.10$).

0.09). No differences were observed in terms of milk fat, lactose, and urea. Nevertheless, SCY had lower SCC compared with CTR at 28 DIM (67.4 vs. $19.6 \pm 17.4 \times 10^3/\text{mL}$ in CTR and SCY, respectively; Trt $\times$ time; $P = 0.04$).

Blood Biomarkers

The least squares means of plasma biomarkers are shown in Table 5. Packed cell volume did not differ between groups. Similarly, markers of energy and protein metabolism did not show any difference between groups. Among inflammation markers, haptoglobin had a reduced increase at calving in the SCY group (Figure 5), resulting in an interaction effect (Trt $\times$ time; $P = 0.03$). Of interest, SCY had lower haptoglobin values both at 1 and 3 DIM (0.37 and 0.45 ± 0.10 g/L, respectively) compared with CTR (0.77 and 0.77 ± 0.10 g/L). Paraoxonase was lower in SCY at 21 and 28 DIM compared with CTR (104.81 vs. 82.00 ± 6.18 U/mL and 107.27 vs. 85.38 ± 6.18 U/mL, respectively), resulting in a tendency toward an interaction effect (Trt $\times$ time; $P = 0.10$). No difference was observed in the other markers of inflamma-

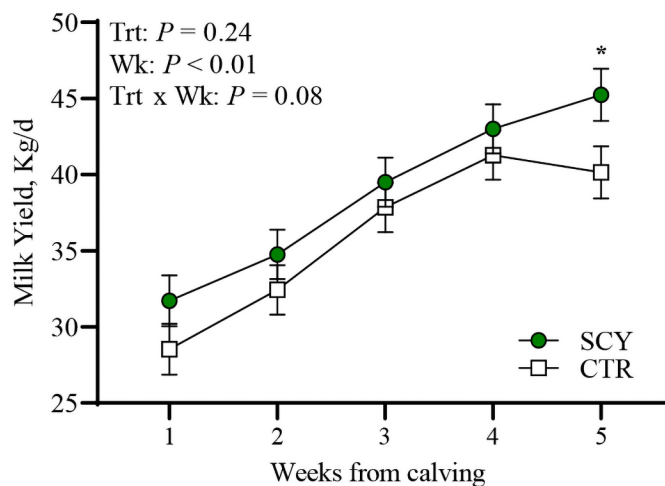


Figure 4. Milk yield in the first 5 wk of lactation in Holstein dairy cows fed a diet not supplemented (CTR) or supplemented with live yeast *Saccharomyces cerevisiae* Sc47 (10 g/d; SCY) from -21 to 21 d relative to calving. Data are presented as LSM $\pm$ SEM. P -values of the main effect are reported in the figure, and differences between groups at each time point are denoted with *. Trt = treatment; Wk = week.

Table 4. Milk composition in the first month of lactation in Holstein dairy cows fed a diet not supplemented (CTR) or supplemented with live yeast *Saccharomyces cerevisiae* Sc47 (10 g/d; SCY) from –21 to 21 d relative to calving

Item	Treatment		SEM ¹	P-value		
	CTR	SCY		Trt ²	Day	Trt × day
Milk fat, %	4.85	4.44	0.39	0.21	0.65	0.46
Milk protein, %	3.38	3.43	0.12	0.75	<0.01	0.09
Milk lactose, %	4.95	5.07	0.11	0.40	<0.01	0.50
Milk urea, mg/dL	25.7	28.5	3.07	0.30	0.42	0.36
Milk SCC, 10 ³ cells/mL	65.9	46.7	17.4	0.18	<0.01	0.04

¹Greatest SEM.²Trt = treatment.

tion. Concentration of IL-1 β tended to increase before calving in SCY and decrease in CTR (Trt × time; $P = 0.06$; Figure 6), whereas IL-6 did not show any difference between groups.

Among liver function markers, GGT increased more after calving in CTR group than in SCY (Trt × Time; $P = 0.03$; Figure 5). No further differences between

groups were observed in terms of markers of liver function, minerals, and antioxidant system, except for a greater magnesemia at 28 DFC in SCY compared with CTR (1.06 vs. 1.20 ± 0.04 mmol/L in CTR and SCY, respectively; $P = 0.04$; Figure 5), and a tendency toward a higher P concentration in SCY (1.67 vs. 1.81 ± 0.05 mmol/L in CTR and SCY, respectively; $P = 0.08$).

Table 5. Plasma biomarkers of Holstein dairy cows fed a diet not supplemented (CTR) or supplemented with live yeast *Saccharomyces cerevisiae* Sc47 (10 g/d; SCY) from –21 to 21 d relative to calving

Biomarker ¹	Treatment		SEM ²	P-value		
	CTR	SCY		Trt ³	Day	Trt × day
PCV, L/L	0.315	0.314	0.004	0.80	<0.01	0.19
Glucose, ⁴ mmol/L	4.02	3.91	0.15	0.64	<0.01	0.21
NEFA, mmol/L	0.75	0.75	0.11	0.97	<0.01	0.22
BHB, mmol/L	0.81	0.73	0.14	0.62	<0.01	0.13
Urea, mmol/L	3.82	3.99	0.27	0.67	<0.01	0.97
Creatinine, μ mol/L	96.3	99.1	4.35	0.66	<0.01	0.68
Cholesterol, mmol/L	2.46	2.59	0.25	0.71	<0.01	0.99
Total protein, g/L	74.6	75.2	2.03	0.82	<0.01	0.35
Albumin, g/L	36.1	36.7	0.78	0.63	<0.01	0.66
Globulin, g/L	38.4	38.6	2.51	0.97	<0.01	0.42
Haptoglobin, g/L	0.37	0.26	0.07	0.28	<0.01	0.03
Ceruloplasmin, μ mol/L	2.63	2.50	0.13	0.48	<0.01	0.69
Paraoxonase, U/mL	81.3	70.1	4.51	0.12	<0.01	0.10
AST-GOT, ⁴ U/L	83.0	86.1	5.75	0.71	<0.01	0.63
GGT, U/L	20.8	20.3	1.26	0.69	<0.01	0.03
Bilirubin, μ mol/L	4.83	4.51	0.98	0.64	<0.01	0.21
Ca, mmol/L	2.49	2.51	0.03	0.65	<0.01	0.29
P, mmol/L	1.67	1.81	0.05	0.08	0.03	0.82
Mg, mmol/L	0.96	1.03	0.04	0.21	<0.01	0.48
Zn, mmol/L	12.8	12.6	0.88	0.83	<0.01	0.78
ROM, mg of H ₂ O ₂ /100 mL	17.6	16.9	0.87	0.61	0.05	0.81
Myeloperoxidase, U/L	420	467	38.9	0.42	0.05	0.48
Thiol groups, μ mol/L	301	317	16.5	0.53	<0.01	0.96
FRAP, μ mol/L	150	153	6.87	0.76	<0.01	0.45
AOPP, μ mol/L	41.5	41.7	1.28	0.91	<0.01	0.81
IL-1 β , pg/mL	26.5	44.4	11.3	0.33	<0.01	0.06
IL-6, pg/mL	360	470	127.7	0.56	<0.01	0.25

¹PCV = packed cell volume; NEFA = nonesterified fatty acids; AST-GOT = aspartate aminotransferase; GGT = γ -glutamyl transferase; ROM = reactive oxygen metabolites; FRAP = ferric reducing ability of plasma; AOPP = advanced oxidation protein products.

²Greatest SEM.³Trt = treatment.⁴Data log-transformed and presented as back-transformed.

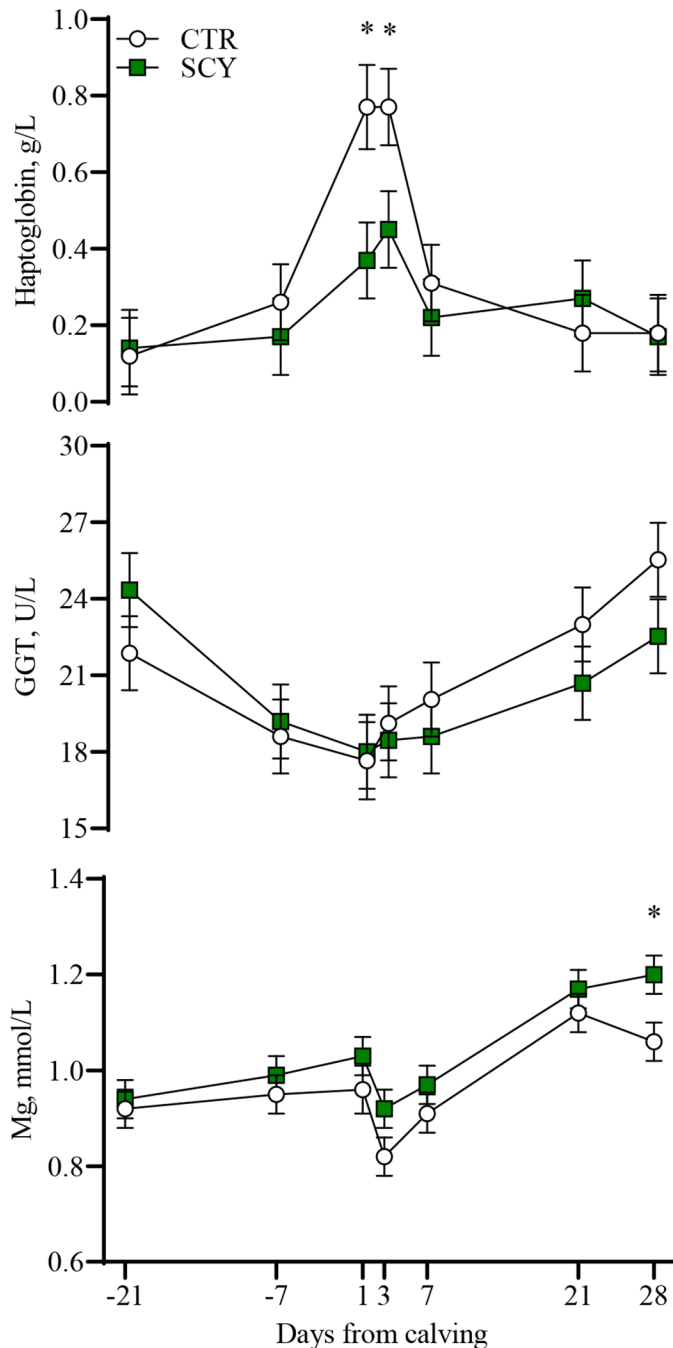


Figure 5. Plasma haptoglobin, γ -glutamyl transferase (GGT), and magnesium from -21 to 28 d relative to calving in Holstein dairy cows fed a diet not supplemented (CTR) or supplemented with live yeast *Saccharomyces cerevisiae* Sc47 (10 g/d; SCY) from -21 to 21 d relative to calving. Data are presented as LSM $\pm$ SEM. Differences between groups at each time point are denoted with *.

DISCUSSION

Several strategies have been tested to improve adaptation to the new lactation in dairy cows. The transition period is indeed the most critical phase of the lacta-

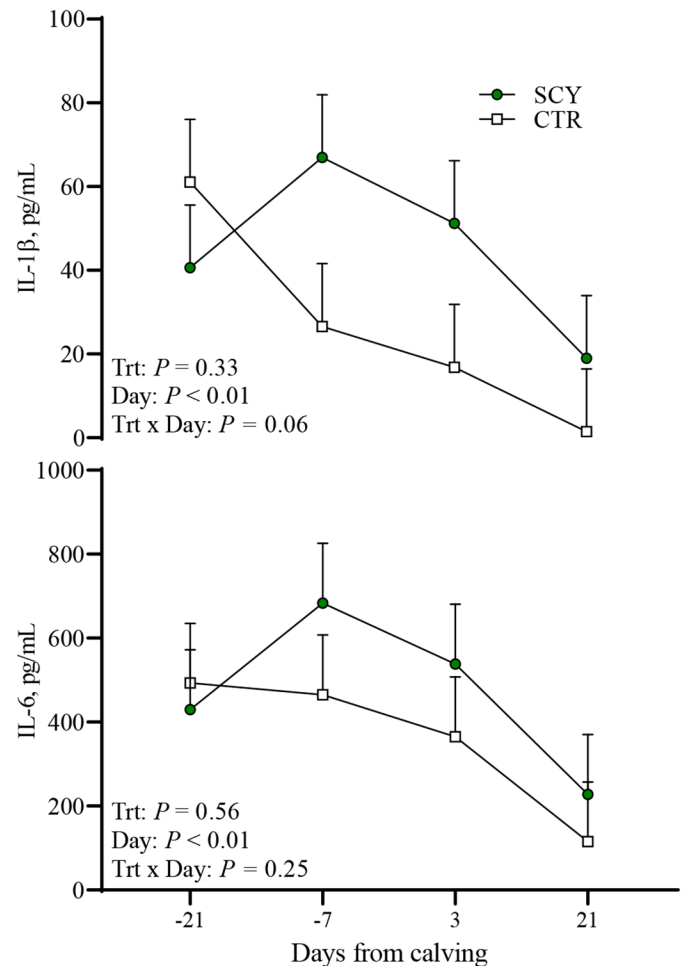


Figure 6. Plasma IL-1 β and IL-6 from -21 to 21 d relative to calving in Holstein dairy cows fed a diet not supplemented (CTR) or supplemented with live yeast *Saccharomyces cerevisiae* Sc47 (10 g/d; SCY) from -21 to 21 d relative to calving. Data are presented as LSM $\pm$ SEM. Trt = treatment; Wk = week.

tion cycle when most of metabolic disorders take place and removal from the herd is greatest (Mezzetti et al., 2021). Most of the nutritional strategies adopted aim to improve rumen functionality in this phase (Bannink et al., 2012). The rumen has a pivotal role in transition period, given the dramatic rise in DMI and diet energy density at calving (particularly due to the increase in highly fermentable feeds) to which it needs to adapt. Among the others, supplementation with live yeasts has been proven to improve rumen fermentation profile, concurrently increasing DMI and milk yield (Desnoyers et al., 2009).

In the present study, performances and markers of rumen function highlighted an overall better condition in SCY cows. In those cows, DMI during the postpartum period was higher. This result is consistent with previous works performed with the same product in

a different stage of lactation cycle (Kumprechtová et al., 2019) or with similar product during the transition period (Bach et al., 2018). However, no difference was found before calving. Effects on DMI following live yeast supplementation are often inconsistent, with some authors reporting positive effects (Desnoyers et al., 2009) and others not finding an effect (de Ondarza et al., 2010). It was proposed that live yeast supplementation may be more effective in stressful conditions (Arambel and Kent, 1990; Moallem et al., 2009), such as, in this case, at the onset of lactation, when the rumen environment undergoes deep modifications. Similar to what happens during heat stress, after calving the risk of low rumen pH and subacute acidosis is high, and the action of yeasts on cellulolytic and lactate-utilizing bacteria may be more effective (Marden et al., 2008; Perdomo et al., 2020). Live *S. cerevisiae* has a scavenging effect on oxygen, which has an inhibitory effect on the growth of the strictly anaerobic bacteria in rumen environment (Newbold et al., 1996), reducing the redox potential and promoting activity of cellulolytic and lactate-utilizing bacteria. Moreover, the improved physiological conditions of the rumen at the onset of lactation in cows receiving SCY can be related to the interaction of some yeast components (e.g., β -glucan fractions) with rumen epithelium. Indeed, it is possible to modulate the inflammation at forestomach (Trevisi et al., 2014, 2018) or gut (Jawhara et al., 2012) epithelia level. In the gut, immune response is orchestrated by the mucosa-associated lymphoid tissue, whereas a nonorganized lymphoid tissue is present in the forestomachs. Molecules such as β -glucan fractions as well as components of microbe walls interact with the host immune system contributing to its local and systemic response (Filipe et al., 2019). Another possible reason for the delayed effect was related to the need for an adaptation period before effects can be observed. Accordingly with improved DMI, daily rumination time was higher in early lactation. This was not surprising because live yeasts act primarily at rumen level. The latter effects were also paired with a higher milk yield, mostly driven by the improved intake and rumen function. As summarized by Moallem et al. (2009), live yeast supplementation positively affects production only when DMI is enhanced, as in the present experiment. Milk protein content tended to be higher in SCY, whereas other studies did not report any effect of SCY on this aspect (Moallem et al., 2009; Kumprechtová et al., 2019). This outcome might be related to a better balance in rumen fermentation, which can result in an increased microbial protein production, eventually available for mammary synthesis (Miller-Webster et al., 2002; Nocek et al., 2011). Of interest, SCC was lower in SCY, fitting within the context of improved inflam-

matory and immune responses (also at the mammary gland level), described thereafter.

Consistent with the aforementioned results, SCY cows had a greater acetate proportion and a lower propionate proportion in the rumen fluid at the peak of rumen fermentation intensity. Live yeasts are known for their positive effect on fiber digestion, achieved by stimulating the activity of cellulolytic bacteria, which produces acetate. Thus, greater acetate:propionate and (acetate + butyrate):propionate ratios were noted in those cows. These ratios are usually considered markers of rumen functionality. Live yeasts promote the activity of cellulolytic bacteria in the rumen environment, improving adhesion of cellulolytic rumen bacteria to cellulose (Roger et al., 1990) and, therefore, improving fiber digestion. The main product of the activity of these bacteria is indeed acetate. Many authors also reported a positive effect on lactate utilization and pH in rumen environment, but we could not detect this effect. The sampling technique we adopted could have affected results because samples collected through an oroesophageal probe are more easily contaminated by saliva than, for instance, rumenocentesis (Duffield et al., 2004). Moreover, results after yeast supplementation are often inconsistent (Desnoyers et al., 2009). Subclinical rumen acidosis can interact with the immune system of the host and induce systemic inflammation (Plaizier et al., 2008, 2022; Zebeli et al., 2015). Indeed, we observed a numerically higher concentration of lactate and lower total VFA in the rumen fluid of SCY cows 21 d after calving. Live yeasts usually decrease both lactate and VFA concentration (Marden et al., 2008). The decrease in total VFA concentration could be due to an increased absorption by rumen epithelium, but the higher lactate, despite not significant, was surprising, also taking into account the other markers, which suggest a better rumen function. In addition, the higher plasma Mg concentration agreed with the aforementioned results because lower levels (in animals fed the same diet) denote a reduced absorption of this mineral by rumen epithelium (Schweigel and Martens, 2000; Minuti et al., 2014), suggesting an impaired rumen function.

Similar to what we observed at the rumen level, live yeast supplementation likely improved hindgut fermentation, as supported by the analysis of fecal VFA, showing increased acetate and decreased propionate proportions. Fecal VFA concentrations are affected by both VFA production in the hindgut and the rate of their absorption. Alterations in hindgut fermentation profile can result in an inflammatory response (Gressley et al., 2011), and the increased epithelium permeability that can be induced by the drastic diet change at the onset of lactation can also contribute to systemic inflamma-

tory states (Eckel and Ametaj, 2016). Therefore, the better fecal VFA profile observed might support the lower inflammatory response noted in SCY cows. Live yeast can survive rumen degradation and throughout the digestive tract (Fomenky et al., 2017), and moderately alter colon gene expression and microbiota (Bach et al., 2018, 2019), confirming another potential effect at this level. Further research on the effect of live yeasts in the intestinal tract are still needed.

The inflammatory response at calving (as shown by lower plasma haptoglobin concentration) was mitigated in SCY group, indicating that inflammatory conditions might also have been implied in the results presented. Inflammatory response is a typical process of transition period (Bradford and Swartz, 2020; Mezzetti et al., 2020). It is widely accepted that all cows experience a certain degree of systemic inflammation at calving, necessary to support physiological adaptations during transition period, but the intensity varies (Bradford et al., 2015). It is challenging to determine whether the improved rumen function improved inflammatory conditions, or whether the reduced inflammation was the driving factor. There is a strong relationship between these 2 pivotal aspects of transition period because rumen dysbiosis can induce systemic inflammation (Garcia et al., 2017). Changes in microbes, microbial metabolism, and the systemic metabolic, hormonal, and immune changes occurring during transition period could alter rumen epithelium, increasing tight junction permeability (Minuti et al., 2015). In turn, loss of epithelial integrity contributes to the development of systemic inflammation (Trevisi et al., 2014; Kvidera et al., 2017). Thus, strategies that are able to improve rumen function (like yeast supplementation), particularly during transition period, can represent a viable way to boost adaptation to the new lactation. Of interest, despite being in the physiological ranges typical of healthy cows (Trevisi et al., 2015), both IL-1 β and IL-6 decreased as DFC increased postpartum but, in SCY IL-1 β concentration (and also IL-6 but not significantly) rose before calving and remained higher than in CTR later. Although understanding the variations in these cytokines around calving is challenging, these decreases have been interpreted as markers of immunosuppression (LeBlanc, 2020; Horst et al., 2021; Brink et al., 2022) or might be related to placenta removal (Vitoratos et al., 2008). Therefore, the different trend in SCY might point a better immune function during transition period in those cows or also a more efficient resolution of the systemic inflammatory response, as supported by the lower haptoglobin. The latter also pointed a milder hepatic acute phase response in SCY cows. Moreover,

consistent with those results, GGT was lower, suggesting a likely better liver function. Liver health during the transition period is strictly connected with inflammatory conditions (Bertoni et al., 2008; Bertoni and Trevisi, 2013). Plasma GGT is used as a marker of liver damage and its lower concentration, paired with the lower haptoglobin, highlighted an overall better condition of SCY cows.

On the other hand, the lower paraoxonase level was unclear. Paraoxonase is usually reduced when liver function is impaired or an inflammatory response takes place (Bionaz et al., 2007). Because other markers of liver function and inflammation were either improved or unaffected, we would tend to exclude this hypothesis in the present study. Nevertheless, this decrease should be further evaluated.

Nevertheless, the limited sample size and the great variability observed might have affected results and prevented the achievement of statistical significance in some of the responses analyzed, which could have provided a more complete framework. Despite that, the results obtained have shown promising effects following live *S. cerevisiae* supplementation, consistent inside the experiment and in line with the findings of previously published studies about comparable products.

CONCLUSIONS

Administration of live *S. cerevisiae* as top feed during transition period (from 21 d before to 21 d after calving) determined an increase in postpartum feed intake and rumination time, paired with a greater milk yield (with a higher protein content) at the onset of lactation. Moreover, it resulted in a better modulation of rumen fermentation, as suggested by the increased acetate and reduced propionate at the peak of postprandial rumen fermentation. Similarly, we found a better modulation of hindgut fermentation, with an increase in acetate proportion during the first week of lactation and a likely improved Mg absorption by rumen epithelium. Furthermore, live yeast-supplemented cows had shortened and milder inflammation around calving (lower rise of plasma haptoglobin concentration and higher IL-1 β concentration) and likely better liver function, as suggested by lower GGT level concentration.

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